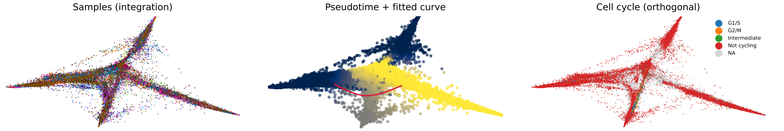
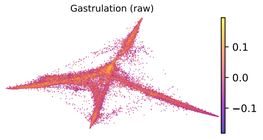
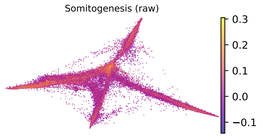
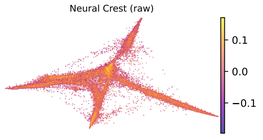
Bassez et al. 2021 — Breast Cancer (10x, 41 samples, 65,519 malignant cells)

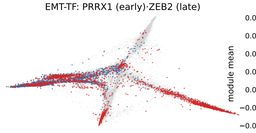
GATE 1 integration / continuity	**GO**	41 samples intermixed on one continuous manifold; sample LISI 6.1→9.1 (of 41) after Harmony θ=10; 363 island cells removed; cycling orthogonal to modules.
GATE 2 module geography	**GO (non-branching)**	Single collapsed axis: pSom/Gast developmental origin → Crest terminal (pSom↔Crest r=−0.82). No Gast arm at any tip; PPT tree over-branches → not a bifurcation.
GATE 3 cross-repression (branching only)	**NA**	Non-branching geometry — no Gast/pSom competing arms to test.

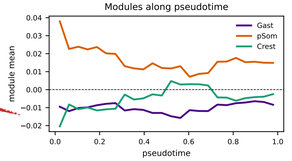













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Parameter choices (derived)
• sample threshold ≥16 (gap-finding) → 41/42 kept
• overdispersed 2498; n_pcs 20; Harmony θ=10 on sample
• n_neighbors 30; UMAP spread 1.3; islands 363 removed
• scFates curve on diffusion rep, Nodes=15,
    λ=0.3, μ=100; fold-free (self-approach 2.56)
• root = pSom/Gast-high tip (pt 0.28); Crest terminal (0.74)
• EMT-TFs PRRX1 (root) · ZEB2 (terminus)
• Annotation: Bassez2021 malignant meta-programs
    (EMT I dominant) + data-derived compartments

Module means (raw): Gast −0.014 · pSom +0.022 · Crest −0.005

VERDICT: pattern reproduces as the non-branching case —
one embryonic axis, pSom/Gast origin → Crest terminus.
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